

# Niayesh Afshordi

Astrophysics, cosmology, and gravity

Professor, University of Waterloo; Research Associate  
Faculty, Perimeter Institute

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Comprehensive public academic CV — updated 2 October 2026

## Profile

Niayesh Afshordi is an astrophysicist working at the interface of cosmology, gravitation, quantum theory, and observation. His research connects fundamental questions to gravitational-wave data, black holes, the cosmic microwave background, large-scale structure, compact objects, and other astrophysical probes. Current themes include quantum black holes and quadratic gravity, strong-gravity tests, early-universe cosmology, dark matter and dark energy, missing baryons, and modified gravity. His interdisciplinary work also includes medical-physics applications, radiotherapy, and epidemiological and public-health modelling.

## Research profiles

- [arXiv author record](#)
- [INSPIRE author record](#)
- [ORCID 0000-0002-9940-7040](#)
- [Google Scholar profile](#)

## Scope

This long public CV retains the complete professional record, including appointments, funding, teaching, supervision, service, presentations, and interviews, while excluding protected personal information. The publication list is refreshed independently from arXiv, INSPIRE, and ORCID. The separate website talks collection holds source decks and slide exports; raw filenames, duplicate export formats, and other speakers' files are not CV entries.

## Professional record

Generated from the Canadian Common CV submitted on 15 August 2025. The protected cover page and personal contact, demographic, citizenship, and identifier fields are excluded. Administrative form metadata is suppressed for readability; the substantive academic record is retained.

Privacy screen: 3 protected source line(s) removed in addition to the protected cover page.

## Language Skills

English and Persian: read, write, speak, understand, and peer review.

## Degrees

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|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2000/9 - 2004/11 | Doctorate, PhD, Astrophysics, Princeton University Thesis Title: The Other 99 Percent Supervisors: Spergel, David, 2000/9 - 2004/11 Research Disciplines: Astronomy and Astrophysics |
| 1996/9 - 1999/5  | Bachelor's Honours, BSc, Physics, Sharif University of Technology                                                                                                                    |

## Recognitions

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| 2020/1          | The Buchalter Cosmology Prize 2019 Dr. Ari Buchalter Prize / Award The \$10,000 First Prize was awarded to Dr. Jahed Abedi of the Max-Planck-Institut für Gravitationsphysik (Albert-Einstein-Institut) and Leibniz Universität Hannover and Dr. Niayesh Afshordi of the University of Waterloo and the Perimeter Institute for Theoretical Physics, for their work entitled "Echoes from the Abyss: A highly spinning black hole remnant for the binary neutron star merger GW170817", published in the Journal of Cosmology and Astroparticle Physics and recognized by the judging panel as "a bold and innovative step towards understanding quantum gravitational phenomena, using observational data to demonstrate gravitational wave echoes from a binary neutron star merger, which could indicate that black hole interiors may be more complex than the simple regions predicted by General Relativity." |
| 2016/1          | The Buchalter Cosmology Prize 2015 Dr. Ari Buchalter Prize / Award The \$2,500 Third Prize was awarded to Dr. Niayesh Afshordi of the University of Waterloo and the Perimeter Institute for Theoretical Physics and Dr. Elliot Nelson of Perimeter Institute, for their work entitled "Cosmological Non-Constant Problem: Cosmological bounds on TeV-scale physics and beyond", published in the Journal of Physical Review D and recognized by the judging panel as "an intriguing proposition that the Planck scale of quantum gravity may soon be accessible by particle accelerators, based on the expected white noise fluctuations in the cosmological constant, which might not be constant at all but vary widely in space."                                                                                                                                                                               |
| 2011/10         | Professor M.K.Vainu Bappu Gold Medal Astronomical Society of India Prize / Award The Vainu Bappu Memorial Gold Medal honours outstanding astronomers worldwide under the age of 35. I was one of the two recipients of this honour in 2011.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 2011/9          | Early Researcher Award - 150,000 (Canadian dollar) Government of Ontario Prize / Award The ERA program helps promising, recently-appointed Ontario researchers build their research teams of undergraduates, graduate students, post-doctoral fellows, research assistants, associates and technicians. The goal of the program is to improve Ontario's ability to attract and retain the best and brightest research talent.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 2010/9 - 2013/9 | Discovery Accelerator Supplement - 120,000 Natural Sciences and Engineering Research Council of Canada (NSERC) Prize / Award The Discovery Accelerator Supplement award provides substantial and timely resources to a small group of researchers whose research proposals suggest and explore highrisk, novel or potentially transformative concepts and lines of inquiry, and are likely to have impact by contributing to groundbreaking advances in the area. I was one of the 8 researchers in Physics who received this award in 2010.                                                                                                                                                                                                                                                                                                                                                                        |

## User Profile

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Researcher Status: Researcher Research Career Start Date: 1998/01/01 Engaged in Clinical Research?: No

**Key Theory / Methodology** General Relativity, Quantum Mechanics, Astrophysical and Cosmological Observations, Numerical Simulations, Large Data, Statistical Methods, Theoretical Modeling

**Research Interests** I lead a comprehensive and ambitious research program, which spans a wide area, from Astrophysics/Cosmology to Fundamental Physics, and includes a diverse spectrum of observational, numerical, and analytic elements. The program aims at linking deep mysteries in fundamental physics to novel observational results and techniques, which can be used to shed light on the structure and laws of our Universe. The areas that my group and I currently focus on are: 1. Phenomenology of Dark Energy and cosmological modifications of General Relativity 2. Distribution of Dark Matter on small scales and novel windows for Dark Matter detection 3. Observational consequences of Aether theories of gravity, and their connection to Quantum Gravity and Cosmological Constant Problem 4. Early Universe, and its links to foundations of Gravity and Quantum Mechanics 5. Missing baryons, and observational probes of intergalactic plasma 6. Physics of Neutron Stars, Black Holes and Accretion Disks.

**Research Experience Summary** I am an Astronomer, and a Physicist; in short I am an Astrophysicist. Currently, I am a Professor in the Astrophysics and Gravitation group at the Department of Physics and Astronomy, University of Waterloo. I am also Research Associate Faculty in the Cosmology and Gravitation group at the Perimeter Institute for Theoretical Physics (PI), the premier Canadian institute for research in theoretical Physics. I finished my PhD at

Princeton University Observatory in August 2004. I was then a fellow at the Institute for Theory and Computation (ITC) at the Harvard-Smithsonian Center for Astrophysics (2004-2007), and a Distinguished Research Fellow at PI (2008-2009). To date, I have published more than 120 peer-reviewed articles, which have been cited more than 7600 times. I have also (co-)supervised 86 students, and 28 postdoctoral fellows. Moreover, I have co-organized 31 international scientific meetings/seminar series, and raised over 1.5 million dollars in external funding.

**Research Specialization Keywords** Astrophysics, Cosmology, Gravitational Waves, Physics of Gravity, Cosmic Microwave Background, Dark Matter, Dark Energy, Black Hole Physics and Astrophysics, Missing Baryons, Lorentz Violation, Quantum Gravity, COVID-19 epidemiology, Flash Radiotherapy

**Disciplines Trained In** Astronomy and Astrophysics, Physics

**Research Disciplines** Astronomy and Astrophysics, Physics, Epidemiology and Biostatistics, Radiology

**Areas of Research** Galactic and Cosmic, Universe Structure, Gravitation, Quantum Phenomena, Elementary Particles

**Fields of Application** Foundations and Knowledge Acquisition, Biomedical Aspects of Human Health, Public Health

## Employment

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| 2022/9          | Full Professor of Physics and Astronomy Physics and Astronomy, University of Waterloo Full-time, Professor Professor in the Astrophysics and Gravitation group. Founding Member of the Waterloo Centre for Astrophysics (2018)                                   |
| 2009/9          | Research Associate Faculty Cosmology and Gravitation, Waterloo, Perimeter Institute for Theoretical Physics A Renewable appointment as a Perimeter Institute for Theoretical Physics Faculty Member.                                                             |
| 2015/9 - 2022/7 | Associate Professor Physics and Astronomy, Waterloo Centre for Astrophysics, University of Waterloo Full-time, Associate Professor Associate Professor in the Astrophysics and Gravitation group. Founding Member of the Waterloo Centre for Astrophysics (2018) |
| 2009/9 - 2015/9 | Assistant Professor Physics and Astronomy, Science, University of Waterloo Full-time, Assistant Professor Assistant Professor in the Astrophysics and Gravitation group                                                                                          |
| 2012/9 - 2013/8 | Visiting Professor Physics, North, SUNY at Buffalo Part-time, Visiting Professorship I visited this department on a regular basis and engaged and collaborated with the faculty in Cosmology.                                                                    |
| 2007/9 - 2009/8 | Distinguished Research Fellow Cosmology and Gravitation, Waterloo, Perimeter Institute for Theoretical Physics                                                                                                                                                   |
| 2004/9 - 2007/8 | Postdoctoral Fellow Astrophysics, Institute for Theory and Computation, Harvard-Smithsonian Center for Astrophysics                                                                                                                                              |
| 2000/9 - 2004/8 | Graduate Student Instructor and Researcher Astrophysical Sciences, Arts and Sciences, Princeton University                                                                                                                                                       |
| 1999/9 - 2000/8 | Graduate Student Teaching Assistant Physics, Arts and Sciences, Brown University                                                                                                                                                                                 |
| 1998/9 - 1999/8 | Research Associate Physics, Tehran, Institute for Theoretical Studies in Physics and Mathematics (IPM)                                                                                                                                                           |
| 1996/1 - 1998/8 | Instructor Iranian National Physics Olympiad Camp, International Physics Olympiad                                                                                                                                                                                |

## Affiliations

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The primary affiliation is denoted by (\*)

(\*) 2015/9 Associate Professor, University of Waterloo

2009/9

## Leaves of Absence and Impact on Research

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2019/9 - 2020/8 Sabbatical, University of Waterloo I was on sabbatical at Perimeter Institute, and did not have any research/teaching duties at U-Waterloo.

## Research Funding History

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### Current

2025/4 - 2030/3 **NSERC Discovery Grant** — *Quantum Black Holes, Vacuum, and the Big Bang: Unraveling Cosmic Mysteries with Multi-Messenger Astronomy*. Principal Investigator. **C\$250,000**.

### Completed

2022/9 - 2025/8 **University of Waterloo Interdisciplinary Trailblazer Grant** — *Quantum Spacetime in the Cosmos: From Conception to Reality*. Principal Investigator; Co-investigator: Doreen Fraser; Collaborator: Steven Weinstein. **C\$80,000**.

2021/9 - 2023/8 **Arthur B. McDonald Institute HQP Pooled Resources Competition**. Principal Investigator. **C\$40,000**.

2020/4 - 2025/3 **NSERC Discovery Grant** — *Novel Astrophysical Windows into Fundamental Physics: From Big Bang to Black Holes, and the Dark Universe*. Principal Applicant. **C\$170,000**.

2015/4 - 2020/3 **NSERC Discovery Grant** — *Novel Astrophysical Windows into Fundamental Physics*. Principal Investigator. **C\$210,000**.

2011/4 - 2016/3 **Ontario Early Researcher Award** — *Astrophysical Windows into Fundamental Physics*. Principal Applicant. **C\$150,000**.

2010/4 - 2015/3 **NSERC Discovery Grant** — *Astrophysical Windows into Fundamental Physics*. Principal Applicant. **C\$280,000**.

2010/4 - 2013/3 **NSERC Discovery Accelerator Supplement** — *Astrophysical Windows into Fundamental Physics*. Principal Applicant. **C\$120,000**.

## Courses Taught

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2025/9 - 2026/4 SCI 238: Introductory Astronomy, University of Waterloo (Fall 2025, Spring 2026, Winter 2026; Winter 2026 co-taught with Mike Hudson)

2013/1 - 2025/4 SCI 238: Introductory Astronomy, University of Waterloo (Winter/Spring 2013, Winter 2015, Winter 2019, Winter/Spring 2021, Spring 2022, Spring 2023, Spring 2025)

2021/9 PHYS 781: Fundamentals of Astrophysics, University of Waterloo (Fall 2021)

2013/9 - 2017/8 PHYS 781: Fundamentals of Astrophysics, University of Waterloo (Fall 2013–Fall 2017)

2018/9 PHYS 475: Astrophysics and Cosmic Structure, University of Waterloo (Fall 2018)

2011/9 PHYS 703: Introduction to Quantum Field Theory, University of Waterloo and Perimeter Institute (Fall 2011)

2010/1 - 2011/4 PHYS 275: Introduction to Astronomy, University of Waterloo (Spring 2010, Spring 2011)

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|--------|---------------------------------------------------------------------------------|
| 2009/9 | Cosmology: Expanding Universe, Perimeter Scholars International (2009)          |
| 2008/9 | Astrophysics and Cosmology through Problems, University of Waterloo (Fall 2008) |

## Student/Postdoctoral Supervision

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### Bachelor's

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| 2023/9 - 2025/8 | Rasheed, Nabeel, University of Waterloo Principal Supervisor Thesis/Project Title: NSERC USRA project: "Analytic models for gravitational wave form templates" Present Position: PhD Student, Massachusetts Institute of Technology                                         |
| 2026/9          | Sahoo, Aliv, University of Waterloo, Undergraduate research intern. Present Position: Undergraduate Student, Indian Institute of Science (IISc)                                                                                                                             |
| 2023/9 - 2024/7 | Yin, Xitai (John), University of Waterloo Principal Supervisor Thesis/Project Title: Black Holes in Quantum Gravity Present Position: MSc Physics – Quantum Technology, University of Waterloo (2025)                                                                       |
| 2023/9 - 2025/8 | Tang, Ruijing, University of Waterloo Principal Supervisor Thesis/Project Title: BSc thesis project: "Exploring Gravitational Effects on accretion disk dynamics in the Teleparallel Born-Infeld gravity" Present Position: Graduate Student, Paris Observatory             |
| 2021/9 - 2023/8 | Liu, Ruolin, University of Waterloo Principal Supervisor Thesis/Project Title: BSc Thesis Project: "Detecting cosmological scalar fields using orbital networks of quantum sensors" Present Position: PhD Student, University of Waterloo and Perimeter Institute           |
| 2022/1 - 2025/4 | Subhi, Hassan, University of Waterloo Principal Supervisor Student Degree Expected Date: 2025/4 Thesis/Project Title: Research Project: "Gravitational Wave Echoes in Pulsar Timing Arrays" Present Position: MMath Student, University of Waterloo and Perimeter Institute |
| 2023            | Kao, Michelle, University of Waterloo Principal Supervisor Undergraduate research co-op in astrophysics and cosmology. Present Position: MSc student in Aerospace Engineering, Carleton University (from Fall 2026)                                                         |
| 2021/2 - 2021/9 | Zheng, Jolene, University of Waterloo Principal Supervisor Thesis/Project Title: Covid forecast modeling and online dashboard Present Position: Software Engineer, Bloomberg                                                                                                |
| 2021/2 - 2021/9 | Olalekan Koiki, Shafika, University of Waterloo Principal Supervisor Thesis/Project Title: Co-op project: "Covid forecast modeling and online dashboard"                                                                                                                    |
| 2021/1 - 2021/8 | Wisniewski, Kasia, University of Waterloo Principal Supervisor Thesis/Project Title: BSc thesis project: "Searching for Black Hole Echoes from the Failed Supernova AT2017be" Present Position: Satellite Operations Technician, GHGSat                                     |

### Bachelor's Honours

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|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2024/9 - 2025/5 | Shah, Saral, University of Waterloo Principal Supervisor Student Degree Expected Date: 2025/5 Thesis/Project Title: Bachelor's thesis: "Dynamical Friction in a neutrinos sea" Present Position: Undergraduate Student, University of Waterloo                             |
| 2024/9 - 2025/5 | Yanover, Lyvia, University of Waterloo Principal Supervisor Student Degree Expected Date: 2025/5 Thesis/Project Title: Bachelor's thesis: "Bi-Thermal Big Bang" Present Position: MSc Student, University of Waterloo                                                      |
| 2024/9 - 2025/5 | Murali, Nikhil, University of Waterloo Principal Supervisor Student Degree Expected Date: 2025/5 Thesis/Project Title: Bachelor's Thesis: "Quantum Black Holes: quantum corrections at the tip of a cigar" Present Position: Undergraduate Student, University of Waterloo |
| 2024/9 - 2025/5 | Brisebois, Alex Dominic Raul, University of Waterloo Principal Supervisor Student Degree Expected Date: 2025/5 Thesis/Project Title: Bachelor's thesis: "Clustering in the Action-Angle Space" Present Position: PhD Student, University of Toronto                        |

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|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2023/9 - 2024/1 | Zhao, Guandi, Fudan University, China Principal Supervisor Thesis/Project Title: BSc visiting student at Perimeter Institute for Theoretical Physics: "Thermal SZ contribution from Low mass galaxies in the unWISE survey" Present Position: MSc student, ETH Zurich, Switzerland                                                    |
| 2023/9 - 2024/8 | Goel, Lakshay, University of Waterloo Co-Supervisor Thesis/Project Title: Accelerated detector in a superposed spacetime Present Position: Co-founder and Team Lead, MedullaAI                                                                                                                                                        |
| 2023/9 - 2024/7 | Gifford, Hazel, University of Waterloo Principal Supervisor Thesis/Project Title: Bohmian Rhapsody: Cuscuton and Cosmology Present Position: MSc Student, Brock University                                                                                                                                                            |
| 2022/9 - 2023/6 | Schiebelbein-Zwack, Aryanna, University of Waterloo Principal Supervisor Thesis/Project Title: BSc Thesis Projects: "Mystery noise in LIGO" Present Position: PhD Student, University of Toronto and CITA                                                                                                                             |
| 2022/1 - 2024/7 | Li, Yu, University of Waterloo Principal Supervisor Thesis/Project Title: NSERC USRA project: "Detecting cosmological scalar fields using orbital networks of quantum sensors" Present Position: PhD Student, California Institute of Technology                                                                                      |
| 2021/5 - 2023/8 | Lawrence, Jensen Pierce Thomas, University of Waterloo Principal Supervisor Thesis/Project Title: Research Project: "Universal horizon formation in the incompressible limit of Einstein-Aether theory" Present Position: PhD Student, Department of Earth, Atmospheric and Planetary Sciences, Massachusetts Institute of Technology |
| 2020/9 - 2023/8 | Wen, Robin, Department of Applied Math, University of Waterloo Principal Supervisor Thesis/Project Title: BSc Thesis project: "A Cosmic Glitch in gravity" Present Position: PhD Student, California Institute of Technology                                                                                                          |
| 2020/9 - 2021/8 | Stewart, Meagan, University of Waterloo Principal Supervisor Thesis/Project Title: BSc thesis project: "Inferring Transient Weak Lensing from Quasar Light Curves" Present Position: MSc Student, Simon Fraser University, British Columbia                                                                                           |
| 2020/5 - 2022/5 | Shawqi, Shafayat, University of Waterloo Principal Supervisor Thesis/Project Title: BSc Thesis Project: "Sorkin-Johnston Effective Field Theory" Present Position: Graduate Research Assistant in Physics, University of Alberta                                                                                                      |
| 2020/1 - 2020/5 | Duff, Petra Mary, University of Waterloo Principal Supervisor Thesis/Project Title: NSERC USRA project: "Echoes from the Abyss" Present Position: PhD Student, University of Toronto                                                                                                                                                  |
| 2019/9 - 2022/8 | Minhas, Palka, University of Waterloo Principal Supervisor Thesis/Project Title: Fast Radio Bursts Present Position: MSc Student, McGill University                                                                                                                                                                                   |
| 2017/1 - 2017/4 | Adair, Connor, University of Waterloo Principal Supervisor Thesis/Project Title: NSERC USRA recipient working on Gravitational Wave Echoes Present Position: MSc Student, University of British Columbia                                                                                                                              |
| 2016/1 - 2016/4 | Dykaar, Hannah, University of Waterloo Principal Supervisor Thesis/Project Title: NSERC USRA research project: "Echoes from the Abyss" Present Position: PhD Student, University of Toronto                                                                                                                                           |
| 2016/1 - 2016/4 | Elgood, Zachary, University of Waterloo Principal Supervisor Thesis/Project Title: NSERC USRA recipient working on Einstein-Aether Black Holes Present Position: La Caixa Doctoral Scholar, Institute of Theoretical Physics (IFT), Madrid, Spain                                                                                     |
| 2010/9 - 2011/6 | Bovard, Luke, University of Waterloo Co-Supervisor Thesis/Project Title: Undergraduate B.Sc. Honours Research Project Present Position: Senior Data Scientist, Plato                                                                                                                                                                  |
| 2008/9 - 2009/8 | Mueller, Matthias, Ulm University Co-Supervisor Thesis/Project Title: Undergraduate Research Project Present Position: Scientist, Forschungszentrum Juelich                                                                                                                                                                           |

## Master's non-Thesis

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|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2026/9 - 2028/8 | Li, Harrison (Haoye), University of Waterloo, Principal Supervisor, MSc student                                                                                                                            |
| 2023/9 - 2025/8 | Liu, Ruolin, University of Waterloo Principal Supervisor Thesis/Project Title: Quadratic Gravity in Big Bang and Black Holes Present Position: PhD Student, University of Waterloo and Perimeter Institute |

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| 2023/3 - present | Friedman-Shaw, Batia, Perimeter Institute for Theoretical Physics/ University of Waterloo<br>Co-supervisor Thesis/Project Title: MSc Project: "Doppler Bias: Impact of Peculiar Velocities on Color Selection and the Large Scale Structure of Galaxy Surveys" Present Position: PhD Student |
| 2022/9 - 2023/6  | Hrytseniak, Oleksandra, Perimeter Institute for Theoretical Physics/ Co-Supervisor University of Waterloo Thesis/Project Title: Project supervision: "Search for flattened non-gaussianity" Present Position: Resident PhD Student, Perimeter Institute and University of Waterloo           |
| 2019/9 - 2020/8  | Chua, Wan Zen, University of Waterloo Principal Supervisor Thesis/Project Title: PSI Research Project: "Electromagnetic Albedo of Fuzzballs" Present Position: PhD student, Cornell University                                                                                               |
| 2015/9 - 2016/6  | Wang, Qingwen, Perimeter Institute for Theoretical Physics Principal Supervisor Thesis/Project Title: Fuzzball approach to black holes Present Position: XVA Quant, Scotiabank                                                                                                               |
| 2013/8 - 2014/6  | Roberts, Brodyn, University of Waterloo/Perimeter Institute for Theoretical Principal Supervisor Physics Thesis/Project Title: PSI Research Project Present Position: Software Engineer, Index Exchange, Toronto, Canada                                                                     |
| 2010/9 - 2012/6  | Agarwal, Abhineet, University of Waterloo Principal Supervisor Thesis/Project Title: Thermal Tachyacoustic Cosmology Present Position: PhD, Jamia Millia Islamia (2020)                                                                                                                      |

## Master's Thesis

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| 2022/9 - 2024/8  | Yang, Jiayue, University of Waterloo Co-Supervisor Thesis/Project Title: Entropy of Analogue Black Holes in Tilted Dirac Cone Materials Present Position: PhD Student, University of Waterloo                                                                                 |
| 2021/1 - 2021/12 | Golafshani, Hamid (Withdrawn), University of Waterloo Principal Supervisor Thesis/Project Title: Topics in Astrophysics and Cosmology Present Position: Professor of Mechanical Engineering, Conestoga College, Waterloo, ON                                                  |
| 2018/9 - 2020/8  | Hergott, Samantha, University of Waterloo Principal Supervisor Thesis/Project Title: Unifying Big Bang and Black Holes Present Position: PhD, York University and Perimeter Institute (2025)                                                                                  |
| 2018/9 - 2019/8  | Yang, Tianyi, University of Waterloo Principal Supervisor Thesis/Project Title: Cosmic Filaments, Galactic Kinematics in Gaia Present Position: PhD Student, University of Edinburgh                                                                                          |
| 2018/9 - 2020/8  | Chen, Alice, University of Waterloo Principal Supervisor Thesis/Project Title: Cosmological Structure Formation Present Position: PhD Student, University of Waterloo                                                                                                         |
| 2017/9 - 2018/7  | Boruah, Supranta, University of Waterloo Co-Supervisor Thesis/Project Title: Constraining the mass of Milky Way based on clustering in the action space of Gaia DR2 stars Present Position: Center for Particle Cosmology Fellow, University of Pennsylvania                  |
| 2016/9 - 2017/8  | Kim, Hyungjin (Tony), University of Waterloo Principal Supervisor Thesis/Project Title: The Early Universe: New aspects of Cuscuton Theory, Modified Dispersion Relations and Pulsar Constraints on Vacuum Noise Present Position: Software Engineer, Validus Risk Management |
| 2014/9 - 2016/8  | Meiers, Michael, University of Waterloo Co-Supervisor Thesis/Project Title: Universal Horizons in Black Holes Present Position: PhD Student, University of California-Davis                                                                                                   |
| 2012/8 - 2013/6  | Okoli, Chiamaka, University of Waterloo/Perimeter Institute for Theoretical Principal Supervisor Physics Thesis/Project Title: Concentration of cosmological haloes Present Position: Deceased                                                                                |
| 2011/9 - 2013/5  | Yazdi, Yasaman, University of Waterloo Principal Supervisor Thesis/Project Title: A Spacetime Approach to Defining Vacuum States and Entropy Present Position: Avadh Bhatia Postdoctoral Fellow, University of Alberta                                                        |
| 2011/8 - 2012/6  | Schuhmann, Robert, University of Waterloo/Perimeter Institute for Principal Supervisor Theoretical Physics Thesis/Project Title: PSI Research Project Present Position: Postdoctoral Research Associate, University of Manchester                                             |

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| 2009/8 - 2010/6  | Ponce, Yosdanis, University of Waterloo/Perimeter Institute for Theoretical Physics Thesis/Project Title: Cosmology of a Black Hole Lattice Present Position: Adjunct Faculty in Physics, Miami Dade College, Wolfson Campus                                                                                                              |
| <b>Doctorate</b> |                                                                                                                                                                                                                                                                                                                                           |
| 2026/9 - 2030/8  | Alharbi, Ohoud K., University of Waterloo, Principal Supervisor, PhD student                                                                                                                                                                                                                                                              |
| 2026/1           | Oberfrank, Robin, Perimeter Institute/University of Waterloo, Co-supervised PhD student                                                                                                                                                                                                                                                   |
| 2023/9 - present | Kolonia, Eleanna (Former Research Assistant), PhD student, Swansea University                                                                                                                                                                                                                                                             |
| 2023/10 - 2025/8 | Nasiri, Arad, Imperial College London Co-Supervisor Student Degree Expected Date: 2025/8 Thesis/Project Title: Project Supervision: "Stochastic Dark Matter: Covariant Brownian Motion from Planckian Discreteness (arXiv:2409.02188)" Present Position: Postdoctoral Fellow, University of New Brunswick                                 |
| 2023/9 - 2024/12 | Patterson, Everett Avison, University of Waterloo Co-Supervisor Student Degree Expected Date: 2025/12 Thesis/Project Title: Project Supervision: "Accelerated detector in a superposed spacetime (arXiv:2409.12441)" Present Position: PhD Student, University of Waterloo                                                                |
| 2023/1 - 2026/12 | Chen, Alice, University of Waterloo and Perimeter Institute for Theoretical Physics Thesis/Project Title: Machine Learning and Cosmic Structure Formation Present Position: PhD Student                                                                                                                                                   |
| 2022/9 - 2025/8  | Borissova, Johanna, Perimeter Institute for Theoretical Physics/ University Co-Supervisor of Waterloo Thesis/Project Title: Scale-invariance at the core of quantum black holes Present Position: Research Assistant, Imperial College London                                                                                             |
| 2022/7 - 2024/12 | Kim, Jungjoon Leo, Queen's University Co-Supervisor Thesis/Project Title: Project Supervision: "A Poisson Log-Normal Framework for Dark Sirens" Present Position: Postdoctoral Fellow, York University                                                                                                                                    |
| 2022/7 - 2022/9  | Boyanov Savov, Valentin, Universidad Complutense de Madrid Principal Supervisor Thesis/Project Title: Visiting PhD student: "Aspects of Quantum Black Holes in semiclassical gravity" Present Position: Postdoctoral Researcher, CENTRA and Instituto Superior Tecnico, Lisbon (through April 2026)                                       |
| 2020/9 - 2024/12 | Dailey, Conner, University of Waterloo and Perimeter Institute Principal Supervisor Student Degree Expected Date: 2024/12 Thesis/Project Title: On the Initial Boundary Value Problem in Numerical Relativity Present Position: Postdoctoral Researcher, Friedrich Schiller University Jena                                               |
| 2019/8 - 2020/4  | Longo, Luis Felipe, University of Sao Paulo, Brazil and Perimeter Institute for Principal Supervisor Theoretical Physics Thesis/Project Title: visiting PhD student: "Echo-Diversity in Binary Black Hole Mergers" Present Position: European Research Council (ERC) Postdoctoral Fellow, Friedrich- Schiller Universitaet, Jena, Germany |
| 2019/2 - 2020/7  | Tsuna, Daichi, University of Tokyo Co-Supervisor Thesis/Project Title: visiting PhD student: "Quantum Black Hole Seismology" Present Position: ITC Fellow, Center for Astrophysics, Harvard & Smithsonian                                                                                                                                 |
| 2018/9 - 2019/8  | Held, Aaron, University of Heidelberg Principal Supervisor Thesis/Project Title: Visiting PhD student: "Quantum Black Holes in Asymptotic Safety" Present Position: Philippe Meyer Junior Research Chair, Ecole Normale Supérieure, Paris                                                                                                 |
| 2018/9 - 2019/8  | Giri, Utkarsh, University of Waterloo/Perimeter Institute for Theoretical Co-Supervisor Physics Thesis/Project Title: Pressure of haloes in cosmological simulations Present Position: Postdoctoral Scholar Research Associate, California Institute of Technology                                                                        |

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| 2018/9 - 2023/8  | Saraswat, Krishan, University of Waterloo/Perimeter Institute for Theoretical Principal Supervisor Physics Thesis/Project Title: Quantum Aspects of Black Holes: From Microstates to Echoes and Somewhere In-Between Present Position: Postdoctoral Fellow, University of Southern California |
| 2018/1 - 2018/9  | Oshita, Naritaka, University of Tokyo Principal Supervisor Thesis/Project Title: visiting PhD student: "Probing microstructure of black hole spacetimes with gravitational wave echoes" Present Position: Assistant Professor, Kyoto University, Yukawa Institute, Kyoto, Japan               |
| 2017/9 - 2018/4  | Dhumuntarao, Aditya, University of Minnesota Principal Supervisor Thesis/Project Title: visiting NSF PhD student: "Temperature for local observers in Quantum Field Theory in curved spacetime" Present Position: Postdoctoral Researcher, Sandia National Laboratories                       |
| 2017/9 - 2021/6  | Robbins, Matthew, University of Waterloo/Perimeter Institute for Theoretical Principal Supervisor Physics Thesis/Project Title: Gravitational Waves, Bose-Einstein Condensates, and Modified Gravity Present Position: Teaching Faculty (tenure-track), University of Toronto                 |
| 2017/9 - 2018/4  | Gherzi, Jose Tomas Galvez, Simon Fraser University Principal Supervisor Thesis/Project Title: visiting PhD student: "Temperature for local observers in Quantum Field Theory in curved spacetime" Present Position: Assistant Professor, Universidad de Ingenieria y Tecnologia, Peru         |
| 2016/9 - 2020/6  | Wang, Qingwen, University of Waterloo/Perimeter Institute for Theoretical Principal Supervisor Physics Thesis/Project Title: Black Hole Echology Present Position: XVA Quant, Scotiabank                                                                                                      |
| 2016/9 - 2018/8  | Abedi, Jahed, Sharif University of Technology Co-Supervisor Thesis/Project Title: Black Hole Echoes Present Position: Postdoctoral Fellow, University of Stavanger, Norway                                                                                                                    |
| 2016/9 - 2017/8  | Haghighi, Mohammad Zhoolideh, Sharif University Principal Supervisor Thesis/Project Title: visiting PhD student: "X-ray Cooling Flows and Modified Gravity" Present Position: Assistant Professor, K. N. Toosi University of Technology                                                       |
| 2015/9 - 2016/8  | Lynch, Morgan, University of Wisconsin-Milwaukee Principal Supervisor Thesis/Project Title: visiting PhD student: "Temperatures of renormalizable quantum field theories in curved spacetime" Present Position: Postdoctoral Researcher, Max Planck Institute for Nuclear Physics             |
| 2014/9 - 2018/6  | Karami, Mansour, University of Waterloo/Perimeter Institute for Theoretical Co-Supervisor Physics Thesis/Project Title: Dark Matter on Small Scales Present Position: Principal AI Engineer and Quantitative Developer, HOOPP                                                                 |
| 2014/9 - 2018/6  | Altamirano, Natacha, University of Waterloo/Perimeter Institute for Co-Supervisor Theoretical Physics Thesis/Project Title: The quantum and the Gravity: Newtonian and Cosmological Applications Present Position: Director, Data Science, Birch Hill Equity Partners, Toronto, Canada        |
| 2014/9 - 2018/6  | Okoli, Chiamaka, University of Waterloo/Perimeter Institute for Theoretical Principal Supervisor Physics Thesis/Project Title: Dark Matter and Neutrinos in the Foggy Universe Present Position: Deceased                                                                                     |
| 2014/4 - 2014/11 | Nelson, Elliot, Penn State University Principal Supervisor Thesis/Project Title: Cosmological Non-Constant Problem Present Position: Future Impact Group                                                                                                                                      |
| 2013/9 - 2017/6  | Yazdi, Yasaman, University of Waterloo Co-Supervisor Thesis/Project Title: Physics and Astrophysics of Black Holes Present Position: Avadh Bhatia Postdoctoral Fellow, University of Alberta                                                                                                  |
| 2013/9 - 2014/8  | Narimani, Ali, University of British Columbia Principal Supervisor Thesis/Project Title: visiting PhD student: "How does pressure gravitate?" Present Position: Google                                                                                                                        |
| 2013/9 - 2014/6  | Karami, Mansour, Sharif University of Technology Co-Supervisor Thesis/Project Title: Cosmic Structure Formation, Modified Gravity and Dark Energy, Gravitational Lensing Present Position: Principal AI Engineer and Quantitative Developer, HOOPP                                            |
| 2013/9 - 2017/6  | Zwane, Nosiphiwo, University of Waterloo/Perimeter Institute for Theoretical Co-Supervisor Physics Thesis/Project Title: Cosmological Tests of Causal Set Phenomenology Present Position: Lecturer, University of Eswatini                                                                    |

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| 2013/9 - 2014/8 | Jeremy Sakstein, University of Cambridge Co-Supervisor Student Degree Start Date: 2010/9 Student Degree Received Date: 2014/7 Thesis/Project Title: visiting PhD student: "Novel Approaches to Modified Gravity" Present Position: Assistant Professor, University of Hawaii                                        |
| 2011/9 - 2017/6 | Gould, Elizabeth, University of Waterloo/Perimeter Institute for Theoretical Principal Supervisor Physics Thesis/Project Title: A Non-Local Reality: Is there a Phase Uncertainty in Quantum Mechanics? Present Position: Postdoctoral Fellow, Arthur B. McDonald Canadian Astroparticle Physics Research Institute |
| 2011/9 - 2016/6 | Saravani, Mehdi, University of Waterloo/Perimeter Institute for Theoretical Principal Supervisor Physics Thesis/Project Title: Aspects of Non-locality Present Position: Executive Director, Goldman Sachs, United Kingdom                                                                                          |
| 2011/9 - 2014/7 | Pourhasan, Razieh, University of Waterloo/Perimeter Institute for Theoretical Co-Supervisor Physics Thesis/Project Title: Adventures in Holography Present Position: Research Assistant, University of Ottawa, Canada                                                                                               |
| 2011/9 - 2015/5 | Kamiab, Farbod, University of Waterloo/Perimeter Institute for Theoretical Principal Supervisor Physics Thesis/Project Title: Neutron Stars, the Exotica - From Modifying General Relativity to Strong Magnetic Fields Present Position: Analytics, Facebook, Dublin, Ireland                                       |
| 2011/1 - 2012/8 | Ponce, Yosdanis (Withdrawn), University of Waterloo/Perimeter Institute for Theoretical Principal Supervisor Physics Student Degree Start Date: 2011/1 Thesis/Project Title: Black Hole Halo Model/ Holographic Cosmology Present Position: Adjunct Faculty in Physics, Miami Dade College, Wolfson Campus          |
| 2010/9 - 2011/9 | Baghran, Shant, Sharif University of Technology Co-Supervisor Thesis/Project Title: The Observational Tests of Dark Energy and Modified Gravity Theories Present Position: Assistant Professor, Sharif University, Tehran, Iran                                                                                     |
| 2010/9 - 2011/6 | Gryb, Sean, University of Waterloo/Perimeter Institute for Theoretical Co-Supervisor Physics Thesis/Project Title: Shape Dynamics and Mach's Principles: Gravity from Conformal Geometrodynamics Present Position: Research Associate, Philosophy Department, University of Bristol, England                        |
| 2010/9 - 2014/6 | Aslanbeigi, Siavash, University of Waterloo/Perimeter Institute for Principal Supervisor Theoretical Physics Thesis/Project Title: Cosmic Atoms: from Causal Sets to Clusters Present Position: Vice President, Trading Products, BMO Capital Markets, Canada                                                       |
| 2008/9 - 2010/6 | Prescod-Weinstein, Chanda, University of Waterloo Co-Supervisor Thesis/Project Title: Cosmic Acceleration As Quantum Gravity Phenomenology Present Position: Assistant Professor, University of New Hampshire                                                                                                       |
| 2007/9 - 2008/8 | Loverde, Marilena, Columbia University Co-Supervisor Thesis/Project Title: Extended Limber Approximation Present Position: Professor and Dr. Ann Nelson Endowed Professor, University of Washington                                                                                                                 |

## Post-doctorate

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| 2024/9 - 2026/8  | Manita, Yusuke, Perimeter Institute, Postdoctoral researcher                                                                                                                                                                                    |
| 2024/3 - 2024/12 | Dunsky, David, New York Univesrity Co-Supervisor Thesis/Project Title: Project Supervision: "Dark Matter Disk Accretion into Black Holes" Present Position: Postdoctoral Fellow, New York Univesrity                                            |
| 2024/3 - 2024/12 | Ghalsasi, Akshay, Harvard University Co-Supervisor Thesis/Project Title: Project Supervision: "Dark Matter Disk Accretion onto Black Holes" Present Position: Postdoctoral Fellow, Department of Physics, Harvard University                    |
| 2024/3 - 2027/3  | Faraji, Shokoufe, University of Waterloo Principal Supervisor Thesis/Project Title: Quantum Gravity and Accretion Physics around Black Holes Present Position: Postdoctoral Fellow, Department of Physics and Astronomy, University of Waterloo |

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| 2023/3 - 2024/12  | Krolewski, Alex, University of Waterloo/ Perimeter Institute for Theoretical Physics Co-Supervisor Thesis/Project Title: Impact of Peculiar Velocities on Color Selection and the Large Scale Structure of Galaxy Surveys Present Position: Research Scientist, California Institute of Technology                                                        |
| 2023/1 - 2025/9   | Quintin, Jerome, University of Waterloo Co-Supervisor Thesis/Project Title: Quadratic Gravity in Cosmology and Black Holes Present Position: Assistant Professor, ETS, and Visiting Fellow, McGill University                                                                                                                                             |
| 2022/8 - 2024/12  | Hassani, Farbod, University of Oslo Principal Supervisor Thesis/Project Title: Project Supervision: "Simulating dynamical friction with cosmic neutrinos" Present Position: Researcher, Institute of Theoretical Astrophysics, University of Oslo                                                                                                         |
| 2022/1 - 2024/12  | Longo, Luis Felipe, Friedrich-Schiller Universitaet, Jena, Germany Principal Supervisor Thesis/Project Title: Spectrogram Correlated Stacking Present Position: European Research Council (ERC) Postdoctoral Fellow, Friedrich- Schiller Universitaet, Jena, Germany                                                                                      |
| 2021/9 - 2022/8   | Muir, Jessica, Perimeter Institute for Theoretical Physics Principal Supervisor Thesis/Project Title: Project supervision: "Search for flattened non-gaussianity" Present Position: Postdoctoral Fellow, Perimeter Institute for Theoretical Physics                                                                                                      |
| 2021/3 - 2025/4   | Mylova, Maria, Perimeter Institute for Theoretical Physics and IPMU, Japan Principal Supervisor Thesis/Project Title: 1- Non-Gaussian Signatures of a Thermal Big Bang 2- Effective Cuscuton Theory Present Position: Project Researcher, Kavli IPMU                                                                                                      |
| 2020/10 - 2024/12 | Nikakhtar, Farnik, Yale University Co-Supervisor Thesis/Project Title: Project Supervision: "Action-Angle Space Clustering" Present Position: YCAA Postdoctoral Fellow, Yale University                                                                                                                                                                   |
| 2020/3 - 2024/3   | Dey, Ramit, Western University, Ontario, Canada Co-Supervisor Thesis/Project Title: 1- Spectrogram correlated stacking: A novel time-frequency domain analysis of the Stochastic Gravitational Wave Background 2- Black Hole Echoes Present Position: Postdoctoral Fellow, Western University, and Associate Postdoctoral Researcher, Perimeter Institute |
| 2020/2 - 2020/6   | Shoshany, Barak, Perimeter Institute for Theoretical Physics Principal Supervisor Thesis/Project Title: Bayesian Search for Black Hole Echoes Present Position: Associate Professor, Brock University                                                                                                                                                     |
| 2019/4 - 2021/3   | Oshita, Naritaka, Perimeter Institute for Theoretical Physics Principal Supervisor Thesis/Project Title: Echoes from Quantum Black Holes Present Position: Assistant Professor, Kyoto University, Yukawa Institute, Kyoto                                                                                                                                 |
| 2019/1 - 2023/12  | Abedi, Jahed, University of Stavanger, Norway Principal Supervisor Thesis/Project Title: Search for Black Hole Echoes in Gravitational Wave Observations Present Position: Assistant Professor, Beijing Institute for Mathematical Sciences and Applications (BIMSA)                                                                                      |
| 2015/9 - 2018/6   | Nelson, Elliot, Perimeter Institute for Theoretical Physics Principal Supervisor Thesis/Project Title: Quantum Decoherence during Inflation from Gravitational Nonlinearities Present Position: Future Impact Group                                                                                                                                       |
| 2013/9 - 2017/8   | Guariento, Daniel, Universidade de Sao Paulo/Perimeter Institute for Principal Supervisor Theoretical Physics Thesis/Project Title: Cosmological Black Holes from Self-Gravitating Fields Present Position: Professor, Conestoga College, Canada                                                                                                          |
| 2013/9 - 2014/8   | Scrimgeour, Morag, University of Waterloo Principal Supervisor Thesis/Project Title: Mass of Neutrinos from Neutrino Wakes Present Position: Data Scientist, Square, San Francisco CA                                                                                                                                                                     |
| 2012/9 - 2015/8   | Marsh, David, Perimeter Institute for Theoretical Physics Thesis/Project Title: Structure of Dark Matter Haloes Present Position: Sofja Kovaleskaja Fellow, University of Goettingen                                                                                                                                                                      |
| 2012/9 - 2016/8   | Kubiznak, David, Perimeter Institute for Theoretical Physics Principal Supervisor Thesis/Project Title: Black Hole Chemistry Present Position: Senior Fellow, Perimeter Scholar International program                                                                                                                                                     |
| 2012/9 - 2016/8   | Hinterbichler, Kurt, Perimeter Institute for Theoretical Physics Principal Supervisor Thesis/Project Title: Consistent modifications of Einstein gravity Present Position: Assistant Professor, Case-Western Reserve University                                                                                                                           |

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| 2012/9 - 2015/8 | Kobayashi, Takeshi, Perimeter Institute for Theoretical Physics/Canadian Principal Supervisor<br>Institute for Theoretical Astrophysics Thesis/Project Title: Primordial Magnetic Fields from the<br>Post-Inflationary Universe Present Position: Associate Professor, SISSA (Scuola Internazionale<br>Superiore di Studi Avanzati) |
| 2011/9 - 2013/8 | Lavaux, Guilhem, University of Waterloo/Perimeter Institute-CITA National Co-Supervisor<br>Fellow Thesis/Project Title: A New Light on Missing Baryons Present Position: CNRS<br>Permanent Position, L'Institut d'Astrophysique de Paris, France                                                                                    |
| 2011/9 - 2012/8 | Baghran, Shant, University of Waterloo/Perimeter Institute for Theoretical Principal Supervisor<br>Physics Thesis/Project Title: Prospects for Detecting Dark Matter Halo Substructure with<br>Pulsar Timing Present Position: Professor at Sharif University, Sharif University, Tehran, Iran                                      |
| 2010/9 - 2013/8 | Zavala, Jesus, University of Waterloo - CITA National Fellow Principal Supervisor<br>Thesis/Project Title: Clustering in the Phase Space of Dark Matter Haloes Present Position:<br>Professor at U-Iceland, University of Iceland                                                                                                   |
| 2010/9 - 2012/8 | Johnson, Matthew, Perimeter Institute for Theoretical Physics Principal Supervisor<br>Thesis/Project Title: Determining the Outcome of Cosmic Bubble Collisions in Full General<br>Relativity Present Position: Associate Professor at York University, Associate Faculty at<br>Perimeter Institute                                 |
| 2009/9 - 2011/8 | Shandera, Sarah, Perimeter Institute for Theoretical Physics Principal Supervisor Thesis/Project<br>Title: Breakdown of Semiclassical Methods in de Sitter Space Present Position: Associate<br>Professor, Penn-State University                                                                                                    |
| 2009/9 - 2013/8 | Erickcek, Adrienne, Perimeter Institute for Theoretical Physics/Canadian Principal Supervisor<br>Institute for Theoretical Astrophysics Thesis/Project Title: Catastrophic Consequences of<br>Kicking the Chameleon Present Position: Assistant Professor, University of North<br>Carolina-Chapel Hill                              |

## Research Associate

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| 2021/9 - 2022/5 | Mukherjee, Suvodip, Perimeter Institute for Theoretical Physics Principal Supervisor<br>Thesis/Project Title: Spectrogram correlated stacking: A novel time-frequency domain analysis<br>of the Stochastic Gravitational Wave Background Present Position: Reader, Tata Institute of<br>Fundamental Research, India |
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## Staff Supervision

Number of Visiting Researchers: 1

## Event Administration

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| 2020/7 - 2024/12 | Member of the Organizing Committee, "Copernicus Weekly Webinar Series" focusing on<br>cosmology and high energy physics <a href="https://indico.cern.ch/event/936284/overview">https://indico.cern.ch/event/936284/overview</a> , Seminar,<br>2020/7 - 2024/12                                                                              |
| 2023/3 - 2024/7  | Local Organizing Committee (Chair), "50 Years of Horndeski Gravity: Exploring Modified<br>Gravity" University of Waterloo and Perimeter Institute for Theoretical Physics, Conference,<br>2024/7 - 2024/7                                                                                                                                   |
| 2019/10 - 2024/6 | Scientific Advisory Committee member, "[RN50] The Event Horizon and Beyond - Celebrating<br>50 Years of Narayan" Boston Museum of Science, Boston, MA <a href="https://www.cfa.harvard.edu/events/rn50-celebrating-50-years-narayan">https://www.cfa.harvard.edu/events/rn50-celebrating-50-years-narayan</a> , Conference, 2024/6 - 2024/6 |
| 2023/11 - 2024/5 | Local Organizing Committee (chair), "McNamara@65: Understanding Feedback in Galaxies and<br>Clusters" University of Waterloo, Conference, 2024/5 - 2024/5                                                                                                                                                                                   |
| 2023/11 - 2024/4 | Scientific Organizing Committee Member, "Cosmological Frontiers in Fundamental Physics"<br>(University of Edinburgh), Conference, 2024/4 - 2024/4                                                                                                                                                                                           |
| 2022/10 - 2023/5 | Organizing Committee (Chair), "Quantum Spacetime in the Cosmos: From Conception to<br>Reality" Perimeter Institute, Conference, 2023/5 - 2023/5                                                                                                                                                                                             |

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| 2020/8 - 2021/5   | Scientific Organizing Committee Member, "Cosmological Frontiers in Fundamental Physics" (Virtual at APC, Paris), Conference, 2021/5 - 2021/5 |
| 2020/2 - 2020/2   | Organizer, "Echoes in Southern Ontario" Perimeter Institute, Workshop, 2020/2 - 2020/2                                                       |
| 2019/5 - 2019/11  | Co-organizer, Everpresent Lambda: Theory meets Observations, Conference, 2019/11 - 2019/11                                                   |
| 2019/3 - 2019/9   | Lead Organizer, "Cosmological Frontiers in Fundamental Physics 2019" Perimeter Institute, Conference, 2019/9 - 2019/9                        |
| 2019/2 - 2019/7   | Chairperson of Parallel Session, 22nd International Conference on General Relativity and Gravitation, Conference, 2019/7 - 2019/7            |
| 2017/9 - 2019/4   | Chair, Astrophysics Seminar Chair, University of Waterloo, Seminar, 2017/9 - 2019/4                                                          |
| 2019/1 - 2019/2   | Scientific Organizing Committee, Tehran Meeting on Cosmology at the Crossroads (Virtual), Conference, 2019/2 - 2019/2                        |
| 2017/9 - 2018/12  | Chair, Physics 10 Undergraduate Lectures, University of Waterloo, Course, 2017/9 - 2018/12                                                   |
| 2018/1 - 2018/6   | Co-organizer, Asymptotic Safety in a Dark Universe, Conference, 2018/6 - 2018/6                                                              |
| 2017/5 - 2017/11  | Co-organizer, Quantum Black Holes in the Sky?, Conference, 2017/11 - 2017/11                                                                 |
| 2016/1 - 2016/7   | Co-organizer, Philosophy of Cosmology, Conference, 2016/7 - 2016/7                                                                           |
| 2015/6 - 2015/12  | Co-organizer, Mann Fest, Conference, 2015/12 - 2015/12                                                                                       |
| 2015/2 - 2015/8   | Co-organizer, Cosmic Flows (and Other Novelties on Large Scales), Conference, 2015/8 - 2015/8                                                |
| 2014/3 - 2014/4   | Co-organizer, Implications of BICEP2, Workshop, 2014/4 - 2014/4                                                                              |
| 2011/5 - 2012/5   | Co-organizer, Conformal Nature of the Universe, Conference, 2012/5 - 2012/5                                                                  |
| 2011/10 - 2012/4  | Co-organizer, Theoretical Physics Workshop, Workshop, 2012/4 - 2012/4                                                                        |
| 2010/9 - 2011/9   | Co-organizer, Unravelling Dark Matter 2011, Conference, 2011/9 - 2011/9                                                                      |
| 2009/12 - 2010/6  | Co-organizer, Cosmological Frontiers of Fundamental Physics, Workshop, 2010/6 - 2010/6                                                       |
| 2008/9 - 2009/8   | Chair, Cosmology Seminar Series, Seminar, 2008/9 - 2009/8                                                                                    |
| 2008/11 - 2009/5  | Co-organizer, New Prospects for Solving the Cosmological Constant Problem, Workshop,                                                         |
| 2009/5            | - 2009/5                                                                                                                                     |
| 2008/5 - 2009/5   | Co-organizer, Sunyaev-Zel'dovich Universe and the Future of Cluster Cosmology, Conference, 2009/4 - 2009/5                                   |
| 2008/10 - 2008/12 | Co-organizer, Young Researchers Conference, Conference, 2008/12 - 2008/12                                                                    |
| 2008/4 - 2008/10  | Co-organizer, PI/CITA Cosmology Mini-Workshop, Workshop, 2008/10 - 2008/10                                                                   |
| 2007/12 - 2008/6  | Co-organizer, Small Scale Structure of Dark Matter, Workshop, 2008/6 - 2008/6                                                                |

2008/1 - 2008/3 Co-organizer, Origins and Observations of Primordial Non-Gaussianity, Workshop, 2008/3 - 2008/3

## Editorial Activities

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2014/1 - 2020/8 Scientific Editor, Research in Astronomy and Astrophysics, Journal

2012/9 - 2013/8 Editorial Board Member, Phys13 News, Newsletter

2010/9 - 2011/8 Editorial Board Member, Phys13 News, Newsletter

## Graduate Examination Activities

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2024/9 - 2024/9 PhD External Examiner, Danieli, Valentina, Astroparticle Physics, Scuola Inter. Superiore di Studi Avanzati

2024/9 - 2024/9 PhD External Examiner, Velluci, Vania, Astroparticle Physics, Scuola Inter. Superiore di Studi Avanzati

## Community and Volunteer Activities

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2009/9 Speaker/Educator, Perimeter Institute for Theoretical Physics I am a regular contributor to outreach events at Perimeter Institute. My responsibilities include giving lectures, participating in Q&A's, and discussions with primary and high school students, as well as high school teachers.

## Knowledge and Technology Translation

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2026/3 - present Consultant, Aligned. Activity Description: Testing frontier AI models in scientific research.

2020/2 - 2023/5 Creator, Research Uptake Strategies Target Stakeholder: General Public Outcome / Deliverable: Based on our study of North American demographic, mobility, climate, and COVID-19 data, we provided a predictive model for growth and mitigation strategies for each US and Canadian city and county. The models were publicly available in the form of an online portal, during April 2021- May 2023. References / Citations / Web Sites: Canada: <http://mylocalcovid.uwaterloo.ca> [ decommissioned in May 2023] US: <https://wolfr.am/COVID19Dash>

2020/4 - 2021/10 Scientific Advisor, Consultation Service Group/Organization/Business Serviced: Public Broadcasting Company (PBS) Target Stakeholder: General Public Outcome / Deliverable: NOVA Universe Revealed: Big Bang, Venture back to the birth of space and time as NOVA probes the mystery of the Big Bang. Premiered: 11/24/21 at 9pm EST Runtime: 53:16, PBS documentary Evidence of Uptake/Impact: While the specific viewership for this program is not publicly released, PBS is the premier public national broadcaster in the US. Each month, approximately 100 million people watch their local PBS stations. References / Citations / Web Sites: <https://www.pbs.org/wgbh/nova/video/nova-universerevealed-big-bang/> Activity Description: I acted as the scientific advisor for the PBS NOVA documentary: "NOVA Universe Revealed: Big Bang", and helped with creating a scientifically accurate portrait of the Big Bang

## International Collaboration Activities

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2019/8 - 2029/8 Member of Laser Interferometer Space Antenna (LISA) consortium, Canada The Laser Interferometer Space Antenna (LISA) is a planned space-based observatory designed to detect and study gravitational waves. As a member of the LISA consortium, I am contributing to the agenda and ideas for the "Fundamental Physics" and "Waveform Modelling" working groups.

## Committee Memberships

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| 2024/7 - 2025/7  | Committee Member, Seminars and Colloquia Committee, Perimeter Institute for Theoretical Physics                                                                                          |
| 2020/9 - 2024/8  | Committee Member, Undergraduate Experience Committee, University of Waterloo Organize and coordinate events to improve student experience                                                |
| 2023/9 - 2024/7  | Committee Member, Postdoctoral Search Committee, Perimeter Institute for Theoretical Physics                                                                                             |
| 2021/9 - 2023/8  | Chair, PhD Student Student Affairs, Perimeter Institute for Theoretical Physics                                                                                                          |
| 2013/1 - 2022/7  | Committee Member, Early Researcher Award Grant Review Panel, Ministry of Research and Innovation, Government of Ontario I served on the review panel for most years, during this period. |
| 2019/9 - 2020/8  | Committee Member, Emmy Noether Fellowship Search Committee, Perimeter Institute for Theoretical Physics                                                                                  |
| 2019/7 - 2019/10 | Chair, "Astrophysics Theory Panel" grant review committee, National Aeronautics and Space Administration (NASA) Chair of a review panel for Theoretical Astrophysics grants              |
| 2014/8 - 2017/8  | Chair, Colloquium Committee, Department of Physics and Astronomy, University of Waterloo                                                                                                 |
| 2013/9 - 2014/8  | Committee Member, Co-op Report Marking Committee, University of Waterloo Ongoing Membership                                                                                              |
| 2010/9 - 2012/8  | Committee Member, Seminar/Colloquium Committee, University of Waterloo                                                                                                                   |
| 2010/9 - 2011/8  | Committee Member, SIN Exam Committee, University of Waterloo                                                                                                                             |
| 2010/9 - 2011/8  | Committee Member, PI/UW Astronomy Faculty Search Committee, University of Waterloo                                                                                                       |
| 2008/9 - 2010/8  | Committee Member, Postdoctoral Recruitment Committee, Perimeter Institute for Theoretical Physics                                                                                        |

## Other Memberships

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| 2021/1          | Member, Canadian Institute for Theoretical Astrophysics, Incorporated (CITA, Inc) The Canadian Institute for Theoretical Astrophysics (CITA), located at the University of Toronto, is a leading research centre dedicated to exploring fundamental questions in theoretical astrophysics. CITA's research spans areas such as cosmology, galaxy formation, black holes, high-energy astrophysics, and planetary science. As a member of CITA Inc., I collaborate with other members across the country to advance theoretical astrophysics. As part of this organization, I contribute to recruiting top talent to Canada and mentoring highly qualified personnel, fostering the next generation of astrophysicists in this dynamic field. |
| 2015/1          | Ordinary Member, Canadian Astronomical Society (CASCA) CASCA is a professional society for Canadian astronomers and researchers involved in the fields of astronomy, astrophysics, and related disciplines. The primary goals of CASCA are to promote the advancement of astronomical research, education, and outreach in Canada.                                                                                                                                                                                                                                                                                                                                                                                                           |
| 2009/9 - 2014/8 | Faculty of Science representative at the Faculty of Math Council, University of Waterloo Ongoing appointment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

## Presentations

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- 1 (2026). *Quantum Gravity in the Wild: Black Hole Echoes, the Big Bang, and a Cosmic Glitch*. Joint Colloquium, ITP/CEICO, Prague, 21 August 2026.
- 2 (2026). *The Natural Log Project: A LISA benchmark for entropy-scale quantum black-hole signatures*. LISA Consortium status update (virtual), 27 July 2026.
- 3 (2026). Quantum Gravity in the Wild: Testing Horizon-Scale Physics with Gravitational Waves and Cosmology. Recent Development of Quantum Gravity and Applications to Cosmology and Black Hole Physics, Hangzhou, China
- 4 (2026). Quadratic Gravity panel contribution. Recent Development of Quantum Gravity and Applications to Cosmology and Black Hole Physics, Hangzhou, China
- 5 (2026). Massive Neutrinos. Perimeter Institute for Theoretical Physics, Waterloo, ON, Canada
- 6 (2026). *Battle of the Big Bang*. Howard University, Washington, DC, United States of America
- 7 (2026). *Battle of the Big Bang*. Kitchener Public Library, Kitchener, ON, Canada
- 8 (2025). Quantum Black Holes & the Conundrum of Cosmic Causality. Physics Colloquium, Arizona State University, Phoenix, AZ, United States of America
- 9 (2025). Battle of the Big Bang: How the Universe Began (and Why It's a Big Mystery!). Physics Literacy for Kids (Zoom talk),
- 10 (2025). Copenhagen Survey on Black Holes and Fundamental Physics. Physics Colloquium (virtual), Tehran, Iran
- 11 (2025). Quantum Black Holes & the Conundrum of Cosmic Causality. Physics & Astronomy Colloquium, McMaster University, Hamilton, ON, Canada
- 12 (2025). Quantum Causality & the Crisis in Physics: Black Holes, the Big Bang, and Beyond. Sydney CPPC virtual seminar, Sydney, Australia
- 13 w/ Stefano Liberati (SISSA). (2024). Quantum Black Holes (concluding panel discussion, and wrap-up). Black Holes inside and out, Niels Bohr Institute, Copenhagen, Denmark
- 14 (2024). Precision Quantum Cosmology: Past, Present, and Future. Gravity and Cosmology 2024, Yukawa Institute for Theoretical Physics, Kyoto University, Kyoto, Japan
- 15 (2024). Quantum Spacetime in the Sky: from the horizon to horizon. Heising-Simons Foundation (HSF) Collaboration on Quantum Gravity and Its Observational Signatures (QuRIOS) Journal Club (virtual), Waterloo, ON, Canada
- 16 (2024). Quantum Black Holes in the Sky. The Event Horizon and Beyond, Celebrating 50 years of Narayan, Boston, MA, United States of America
- 17 (2024). Quantum Black Holes & the Conundrum of Cosmic Causality. Physics Colloquium, International Center for Theoretical Physics Asia-Pacific (ICTP-AP), Beijing, China
- 18 (2024). Quantum Spacetime in the Sky: from the horizon to the vacuum. Bottom-Up Cross-Cutting Workshop "JENAS Initiative: Gravitational Wave Probes of Fundamental Physics" 12–16 Feb 2024 Sapienza University of Rome, Rome, Italy
- 19 (2024). Quantum Black Holes & the Conundrum of Cosmic Causality. Physics Colloquium, Institute of High Energy Physics, Beijing, China
- 20 (2024). Quantum Spacetime in the Sky: from the horizon to horizon. International Congress of Basic Science 2, Beijing Institute for Mathematical Science and Applications (BIMSA), Beijing, China
- 21 (2024). Imaging Quantum Black Holes & (Towards) Optimal Estimators of Stochastic Gravitational Wave Background. The 1st Tsinghua Relativity Workshop, Beijing, China

- (2023). The Big Picture: New Vistas in Gravitational Wave Astronomy. Canadian Astronomical Society (CASA) Annual General Meeting, Penticton, BC, Canada
- (2023). Quantum Spacetime and the Cosmic Endgame. Physics Colloquium, Simon Fraser University, Burnaby, Canada
- (2023). A Tale of Stars and Sirens. The Past, Present and Future of Astronomical Surveys - A Celebration of the Scientific Careers of Michael Strauss and Robert Lupton, Princeton University, Princeton, NJ, United States of America
- (2023). Quantum Gravity and the Cosmos. Testing Aspects of General Relativity-II (virtual), Waterloo, ON, Canada
- (2023). The paradox of human and the cosmos. Ottawa Intellectual Discussions (Virtual), Waterloo, ON, Canada
- (2023). The Endgame: Quantum Spacetime and the Cosmos. Physics Colloquium, University of British Columbia, Vancouver, BC, Canada
- (2023). The Big Picture. DE Shaw Company (virtual seminar), Waterloo, ON, Canada
- (2023). Quantum Spacetime in the Sky: from Conception to Reality. ISRA 2023: The International Symposium on Recent Developments in Relativistic Astrophysics, Gangtok, India
- (2023). Into the Unknown: A Journey on the Dark Side. Virtual Public Lecture for Lord Byng high school Astronomy Club (Vancouver, BC), Waterloo, ON, Canada
- (2023). From Black Holes to the Big Bang: Humanity's Cosmic Paradox. Public Lecture at Kitchener Public Library, Kitchener, ON, Canada
- (2023). Spectrogram correlated stacking: A novel time-frequency domain analysis of the Stochastic Gravitational Wave Background. Cosmology and Gravitation Seminar, Perimeter Institute, Waterloo, ON, Canada
- (2023). Quantum Precision Cosmology. Testing Gravity 2023, Vancouver, BC, Canada
- (2023). The War Within: Humanity's Cosmic Paradox. Physics 10 seminar series, University of Waterloo, Waterloo, ON, Canada
- (2022). Einstein vs Einstein. Testing Aspects of General Relativity (Joint Canadian/Indian virtual conference), Waterloo, Canada
- (2022). Einstein vs Einstein: The Great War. Physics Colloquium Institute for Research in Fundamental Sciences (IPM), (remote presentation), Tehran, Iran
- (2022). Mapping the Big Bang. Remote Public Talk at Farzanegan High School, Tehran (virtual talk in Persian), Tehran, Iran
- (2022). Stimulating Hawking Radiation of Gravitational waves: from theory to observation. New horizons for (no-)horizon physics: from gauge to gravity and back, Galileo Galilei Institute, Florence, Italy
- (2022). Einstein vs Einstein: The Great War. Cosmo Coffee Seminar Series, Theory Group, the European Organization for Nuclear Research (CERN) (remote presentation), Geneva, Switzerland
- (2022). Stimulating Hawking Radiation of Gravitational Waves. Quantum Simulators of Fundamental Physics (virtual seminar series), Waterloo, Canada
- (2022). Quantum Black Holes: from holography to scrambling and gravitational wave astronomy. The 1st International Conference of Holography and its Applications, Damghan University (remote presentation), Damghan, Iran
- (2022). Fundamental Physics vs. Fundamentalism. History, Philosophy, and Sociology of Cosmology and Astroparticle Physics Conference, Bonn, Germany
- (2021). Cosmology of the Pandemic. Journal-club Galaxies, Institut d'Astrophysique de Paris (remote talk), Paris, France

- 44 (2021). Einstein vs Einstein: The Great War. Gravity Seminar Series, University of Southampton (remote presentation), Southampton, United Kingdom
- 45 (2021). Black Holes, Unscripted. Copernicus Webinar Series (virtual presentation), Waterloo, Canada
- 46 (2021). On the origin of the LIGO "mystery" noise and the high energy particle physics desert. Sixteenth Marcel Grossmann Meeting (virtual), Waterloo, ON, Canada
- 47 (2021). Physics and Astrophysics with Gravitational Waves. 14th Iranian national conference on Astronomy and Astrophysics, Semnan University (Virtual public talk in Persian), Semnan, Iran
- 48 (2021). Big Bang: Fact vs Fiction. The 28th IPM Physics Spring Conference, Tehran, Iran (remote presentation), Tehran, Iran
- 49 (2021). Cosmology and the Pandemic. Virtual conference on Gravity and Cosmology, Shahid Beheshti University (Remote keynote presentation in Persian), Tehran, Iran
- 50 (2021). A New Way to Measure Mass. Streams 21 (virtual conference), Waterloo, Canada
- 51 (2021). Cosmology: Triumphs, Trials, and Tribulations. Physics Colloquium, Tarbiat Modares University (remote presentation), Tehran, Iran
- 52 (2021). From Cosmos to the Pandemic. 37th Science and Technology Conference, Allameh Helli High School (virtual talk in Persian), Tehran, Iran
- 53 (2021). Into the Unknown. Physics Colloquium, University of Northern Iowa (remote), Cedar Falls, IA, United States of America
- 54 (2021). Black Holes, Unscripted. Black-Hole Microstructure, virtual conference organized by U-Saclay, France (remote presentation), Saclay, France
- 55 (2021). From Cosmology to the Pandemic. Sharif University of Technology Alumni Associate Webinar Series (virtual talk in Persian), Waterloo, ON, Canada
- 56 (2021). On Nothing!. Alternative Gravities and Fundamental Cosmology 2021 (virtual conference, organized by University of Szczecin, Poland), Szczecin, Poland
- 57 (2021). On the origin of the LIGO "mystery" noise and the high energy particle physics desert. The XXVIII International Conference on Supersymmetry and Unification of Fundamental Interactions (SUSY 2021, virtual), Waterloo, ON, Canada
- 58 (2021). From the Cosmos to the Pandemic: A Dive into the Unknown. 26th Special Physics School, Institute for Advanced Studies in Basic Sciences (remote talk in Persian), Zanjan, Iran
- 59 (2020). Probing Physics Beyond the Standard Model with the Gravitational Wave "Mystery" Noise. Quantum Spacetime and the Renormalization Group (organized by CP3 Origins, Denmark, virtual conference), Waterloo, ON, Canada
- 60 (2020). Local drivers of growth and containment for COVID-19 pandemic. Seminar Series on the Mathematical Modelling of COVID-19, Fields Institute for Research in Mathematical Sciences, Toronto, Canada
- 61 (2020). A New Way to Measure Mass. CANadian Virtual Astronomy Seminar (CANVAS) (virtual presentation), Waterloo, Canada
- 62 (2020). On the Edge of the Abyss. Virtual Seminar, McGill Space Institute, Montreal, QC, Canada
- 63 (2020). A New Way to Measure Mass. Galactic Dynamics Journal Club, University of Pennsylvania (remote presentation), Philadelphia, PA, United States of America
- 64 (2020). Quantum Black Hole Seismology. The 13th International LISA Symposium (virtual conference, prerecorded talk), Waterloo, Canada
- 65 (2020). Quantum Black Hole Seismology. 23rd meeting in Astronomical Research, Zanjan, Iran (remote presentation), Zanjan, Iran

- 66 (2020). A novel predictive framework for the novel coronavirus. Centre for Disease Modelling, Canada- China Distinguished Lecture, Canada
- 67 (2020). Quantum Black Holes in the Sky. Quantum Gravity and the Renormalization Group, Odense, Denmark
- 68 (2020). Managing the COVID-19 Pandemic across Geography and Demography. seminar for the Google COVID-19 data working group (virtual), Waterloo, ON, Canada
- 69 (2020). On the secrets of black holes, worm holes, white holes, and the firewalls. Remote public talk in Persian for Astronomy (Nojoom) Magazine, Tehran, Iran
- 70 (2020). Quantum Black Hole Seismology. Astrophysics Seminar Series, Bar-Ilan University, Israel (remote presentation), Israel
- 71 (2020). Quantum Black Hole Seismology. High Energy Physics (HEP) seminar series, school of physics, Institute for Research in Fundamental Sciences (IPM), Tehran (remote presentation), Tehran, Iran
- 72 (2020). Managing the COVID-19 Pandemic across Geography and Demography. Seminar Series, Astronomy School, Institute for Research in Fundamental Sciences (IPM), Tehran (remote, tehran, Iran
- 73 (2020). A New Way to Measure Mass. Astrophysics Seminar, Sharif University (remote presentation), Tehran, Iran
- 74 (2020). The invisible 99 percent. Physics Club (remote public talk in Persian), Tehran, Iran
- 75 (2020). Managing the COVID-19 Pandemic across Geography and Demography. Colloquium, Perimeter Institute, Waterloo, ON, Canada
- 76 (2020). Hubble Tension and an Everpresent Lambda. 3rd Online Workshop on Quantum Gravity, “Quantum Gravity meets dark energy” (virtual conference), waterloo, Canada
- 77 \*Aslanbeigi, S; \*Saravani, M. (2019). Dark Matter from Quantum Gravity. Everpresent Lambda: Theory Meets Observations, Perimeter Institute for Theoretical Physics, Waterloo, ON, Canada
- 78 (2019). Dark Energy [On Nothing]. CosmoCruise 2019, Venice, Italy
- 79 (2019). A Tale of Big Bang. Milky Way Persian Science Channel, Iran (virtual), Waterloo, ON, Canada
- 80 (2019). Quantum Black Holes in the Sky. Testing Gravity 2019, Vancouver, BC, Canada
- 81 (2019). Cosmological non-Constant (CnC) Problem. 5th Bahamas Advanced Study Institute and Conferences 2019 (BASIC 2019) (remote presentation), Bahamas
- 82 (2019). Quantum Black Holes in the Sky. Testing Gravity 2019, Vancouver, Canada
- 83 (2019). On Nothing!. Physics Colloquium at Grand Valley State University, Grand Rapids, MI, United States of America
- 84 (2018). Quantum Black Holes in the Sky: From Quantum Gravity to Astrophysics and Cosmology. Solvay-Perimeter-APC Workshop on "Cosmological Frontiers in Fundamental Physics" - Solvay Institute, Brussels, Belgium
- 85 (2018). Quantum Black Holes in the Sky: from Quantum Gravity to Astrophysics and Cosmology. Cosmology and Gravitation Seminar Series, Perimeter Institute, Waterloo, ON, Canada
- 86 (2018). Seeing through Simulation Blindspots. Rotman Summer Institute on Philosophy of Cosmology, Goderich, Canada
- 87 (2018). Reviving Aether in Einstein’s Universe: Trials, Tribulations, and Triumphs. Lecture for undergraduate Cosmology and Gravitation class at Western University, London, ON, Canada
- 88 (2018). Seeing Cosmic Nanostructure: From Plasma to Dark Matter. Seminar - Canadian Institute for Theoretical Astrophysics, University of Toronto, Toronto, Canada

- 89 (2018). Quantum Black Holes in the Sky: from Quantum Gravity to Astrophysics and Cosmology. Physics and Astronomy Colloquium, Western University, London, ON, Canada
- 90 (2018). Quantum Black Holes in the Sky: from Quantum Gravity to Astrophysics and Cosmology. Theoretical High Energy Physics Seminar Series, McGill University, Montreal, QC, Canada
- 91 (2018). Adios, Lorentz! Hola, Kuhn!. Rotman Summer Institute on Philosophy of Cosmology, Goderich, Canada
- 92 (2018). Cosmological non-Constant Problem. Asymptotic Safety in a Dark Universe, Perimeter Institute, Waterloo, ON, Canada
- 93 (2018). Quantum Black Holes in the Sky: From Quantum Gravity to Astrophysics and Cosmology. HEP Seminar - McGill University, Montreal, Canada
- 94 (2018). The Other 99 Percent: A Dark Cosmos. Keynote lecture for high school teachers, Perimeter Institute, Waterloo, ON, Canada
- 95 (2018). Quantum Black Holes in the Sky: from Quantum Gravity to Astrophysics and Cosmology. Cosmology and Gravitation Seminar, Perimeter Institute, Waterloo, ON, Canada
- 96 (2017). Echoes from the Abyss. Testing Gravity 2017, Vancouver, Canada
- 97 (2017). Reflections on Spacetime. Cosmology and the Future of Spacetime - Western University, London, Canada
- 98 (2017). Echoes from the Abyss. Seminar - Department of Applied Mathematics, University of Waterloo, Waterloo, Canada
- 99 (2017). Quantum Black Holes in the Sky. Colloquium - University of Kansas, Lawrence, Kansas, United States of America
- 100 (2017). Quantum Black Holes in the Sky. Colloquium - Department of Physics, Ohio State University, Columbus, United States of America
- 101 (2017). Quantum Black Holes in the Sky - Plenary Lecture. CosPA 2017, Kyoto, Japan
- 102 (2017). How to turn Quantum Gravity into Science?. Probing the Spacetime Fabric: From Concepts to Phenomenology - SISSA, Trieste, Italy
- 103 (2017). Echoes from the Abyss. Physics and Astrophysics at the eXtreme (PAX), Amsterdam, Netherlands
- 104 (2017). Seeing Dark in the Extremes: From Nanostructure, to Infinity and Beyond. CPARC Colloquium - Queen's University, Kingston, Canada
- 105 (2016). The Many Faces of the Cosmological Constant Problem. Experimental Search for Quantum Gravity - Frankfurt Institute for Advanced Studies, Frankfurt, Germany
- 106 (2016). Cosmological Non-Constant Problem: Astrophysical Limits of Particle Physics. HET-Discovery Seminar - Niels Bohr Institute, University of Copenhagen, Copenhagen, Denmark
- 107 (2016). E.F.T. is Evil!. Unifying Tests of General Relativity - Walter Burke Institute for Theoretical Physics, Caltech, Pasadena, United States of America
- 108 (2016). Accretion in Radiative Equipartition. Seminar - Dark Cosmology Centre, University of Copenhagen, Copenhagen, Denmark
- 109 (2015). (Real) Cosmology with Quark-Gluon Plasmas. Cosmology/HEP Seminar - Brookhaven National Laboratory, Upton, New York, United States of America
- 110 (2015). Pulsar Timing Constraints on Physics Beyond Standard Model. Mann Fest - University of Waterloo, Waterloo, Canada
- 111 (2015). Cosmological (Non-)Constant Problem. Testing Gravity 2015, Vancouver, Canada

- (2015). First Erlangen Workshop on Quantum Gravity and Cosmology. High Energy Physics Seminar -  
112 University of Toronto, Toronto, Canada
- (2015). Cosmological Non-Constant Problem: The Case for TeV Scale Quantum Gravity. First Erlangen  
113 Workshop on Quantum Gravity and Cosmology, Erlangen, Germany
- (2015). Alice in the (Relativistic) Wonderland. Colloquium - Department of Mathematics, Western University,  
114 London, Canada
- (2014). Probing Dark Matter Nanostructure. Mitchell Workshop on Collider and Dark Matter Physics, College  
115 Station, United States of America
- (2014). From Black Holes to Big Bang, and Back!. Cosmology Seminar - Perimeter Institute for Theoretical  
116 Physics, Waterloo, Canada
- (2014). Cosmological (Non)-Constant Problem. Large Angle Anomalies: from a posteriori to a priori - Case-  
117 Western Reserve University, Cleveland, United States of America
- (2014). An Introduction to the Universe - Plenary Talk. Sharif University Alumni Conference, Tehran, Iran  
118
- (2014). From Black Holes to Big Bang, and Back!. Colloquium - Department of Physics, Isfahan University of  
119 Technology, Isfahan, Iran
- (2014). Probing Dark Matter Nanostructure. Cosmology Seminar - Department of Physics, Sharif University of  
120 Technology, Tehran, Iran
- (2014). Dark Matter Nanostructure. Astrophysics Seminar - Department of Physics and Astronomy, University  
121 of Waterloo, Waterloo, Canada
- (2014). From Black Holes to Big Bang, and Back!. Invited Colloquium - School of Astronomy, Institute for  
122 Research in Fundamental Sciences (IPM), Tehran, Iran
- (2014). An Introduction to the Universe, Q&A. Zaferanieh Amateur Observatory, Tehran, Iran  
123
- (2014). Life after Lorentz: Quantum Mechanics, Gravity, and the Crisis of Falsifiability. Physics Colloquium -  
124 University of Tehran, Tehran, Iran
- (2013). Missing Baryons: How Much? How Hot? How Fast?. Cross-Correlating Cosmic Fields Conference,  
125 Shanghai, China
- Gould E. (2013). A Non-Local Reality: Is there a Phase Uncertainty in Quantum Mechanics?. Quantum  
126 Foundations Seminar - Perimeter Institute for Theoretical Physics, Waterloo, Canada
- (2013). Tales of Dark Matter on Small Scales. Colloquium - Kavli Institute for Astronomy and Astrophysics,  
127 Beijing, China
- (2013). Dynamical Emergence of Universal Horizons During the Formation of Black Holes. LOOPS 13,  
128 Waterloo, Canada
- (2013). Out of the White Hole: A Holographic Origin for the Big Bang. Cosmology After Planck 2013, Ann  
129 Arbor, United States of America
- (2013). Where Will Einstein Fail?. CASCA 2013, Vancouver, Canada  
130
- (2013). Why is Particle Physics Lorentz-Invariant?. Kavli Workshop - IPMU Focus Week on Gravity and  
131 Lorentz Violation, Kashiwa, Japan
- (2013). Life after Lorentz: Quantum Mechanics, Gravity, and the Crisis of Falsifiability. Colloquium -  
132 University of Wisconsin-Madison, Madison, United States of America
- (2013). Tales of Dark Matter on Small Scales. TAMU Workshop on Dark Matter, College Station, United  
133 States of America
- (2012). Horava-Lifshitz Cosmology and Dark Matter Holes - 2-Lecture Series. Cape Town International  
134 Cosmology School 2012, Stellenbosch, South Africa, Republic of

- (2012). Hot Big Bang at a Lifshitz Point. CAP Congress 2012, Calgary, Canada  
135
- (2012). Reviving Gravity's Aether in Einstein's Universe. Colloquium - York University, Toronto, Canada  
136
- (2012). Tales of Dark Matter on Small Scales. Theoretical Physics Workshop - UWO, London, Canada  
137
- (2012). Why is Particle Physics Lorentz Invariant?. Experimental Search for Quantum Gravity: The Hard  
138 Facts, Waterloo, Canada
- (2012). Modern Cosmology. Cosmology & Gravitation Seminar - Perimeter Institute for Theoretical Physics,  
139 Waterloo, Canada
- Bonanno A, Brandenberger R, Magueijo J, Nemiroff R, Amelino-Camelia G. (2012). Photons and Quantum  
140 Gravity in Astrophysics. Experimental Search for Quantum Gravity 2012, Waterloo, Canada
- (2012). Trials and Tribulations of Aether. Conformal Nature of the Universe Conference, Waterloo, Canada  
141
- (2011). Where will Einstein Fail? Insights into Gravity and Dark Energy. CITA Invited Seminar - University of  
142 Toronto, Toronto, Canada
- (2011). Largest and Smallest Dark Matter Clusters, and What We Can Learn from Them. High Energy Theory  
143 Seminar - University of Michigan, Ann Arbor, United States of America
- (2011). Structure in the Phase Space and Dark Matter Astronomy. Unravelling Dark Matter Conference 2011,  
144 Waterloo, Canada
- (2011). Where will Einstein Fail? Insights into Gravity and Dark Energy. KICP Colloquium - University of  
145 Chicago, Chicago, United States of America
- (2011). Aether, Quantum Gravity and Cosmology. Philosophy of Cosmology Workshop - 11th Annual  
146 Conference in the Foundations of Physics, London, Canada
- (2011). Where will Einstein Fail? Insights into Gravity and Dark Energy. Vainu Bappu Memorial Gold Medal  
147 Award Ceremony - Astronomical Society of India, Pune, India
- (2011). Aether, Quantum Gravity and Cosmology. Cosmological Frontiers in Fundamental Physics 2011, Paris,  
148 France
- (2011). How can Astrophysical Compact Objects Shed Light on Dark Cosmology?. High Energy Theory  
149 Seminar - University of Amsterdam, Amsterdam, Netherlands
- (2011). Fine Structure of Dark Matter Phase Space. Dark Matter: Its Origin, Models, and Detection,  
150 Albuquerque, United States of America
- (2010). Non-Linear Structure Formation in Cosmology - 2-Part Lecture Series. Invited Lectures on Structure  
151 Formation - Institute for Research in Fundamental Sciences, Tehran, Iran
- (2010). First Precision Measurement in Quantum Gravity. Cosmology Seminar - Institute for Research in  
152 Fundamental Sciences, Tehran, Iran
- (2010). Stellar Black Holes and the Thermodynamic Origin of Cosmic Acceleration. Colloquium - Institut  
153 d'Astrophysique de Paris, Paris, France
- (2010). Cosmological Constant Problem: Rethinking Quantum and Gravity. Laws of Nature: Their Nature and  
154 Knowability 2010, Waterloo, Canada
- (2010). Cosmology and the End of Physics!. Colloquium - University of Guelph, Guelph, Canada  
155
- (2010). Hierarchy in the Phase Space and Dark Matter Astronomy. Astrophysics Seminar - Queen's University,  
156 Kingston, Canada
- (2009). The End of the Cosmological Constant Problem. Astrophysics Workshop - Interactions in the Dark  
157 Sector, Leiden, Netherlands

- 158 (2009). Black Hole Horizons and the Origin of Cosmic Acceleration. Holographic Cosmology Workshop 2009, Waterloo, Canada
- 159 (2009). Gravitational Aether as a Solution to the CC Problem and more!. New Prospects for Solving the Cosmological Constant Problem 2009, Waterloo, Canada
- 160 (2009). Stellar Black Holes and the Origin of Cosmic Acceleration. Cosmology Seminar - KIPAC, Stanford, United States of America
- 161 (2009). Hierarchy in the Phase Space and Dark Matter Astronomy. CYGNUS 2009, Cambridge, United States of America
- 162 (2009). The End of the Cosmological Constant Problem. Cosmology Seminar - IPMU, Kashiwa, Japan
- 163 (2009). Stellar Black Holes and the Origin of Cosmic Acceleration. Theory Canada 5 Conference, Fredericton, Canada
- 164 (2009). Cuscuton and Solving the Cosmological Constant Problem. High Energy Physics Seminar - Vanderbilt University, Nashville, United States of America
- 165 (2009). Cuscuton and Horava-Lifshitz Gravity. Gravity at a Lifshitz Point 2009, Waterloo, Canada
- 166 (2009). (Towards) the End of the Cosmological Constant Problem!. Cosmology & Gravitation Seminar - Perimeter Institute for Theoretical Physics, Waterloo, Canada
- 167 (2008). Gravitational Aether as a Solution to the Cosmological Constant Problem. 14th International Symposium on Particles, Strings and Cosmology (PASCOS '08), Waterloo, Canada
- 168 (2008). Physics Beyond the Horizon. Cosmology and Gravitation Seminar - Perimeter Institute for Theoretical Physics, Waterloo, Canada
- 169 (2008). Physics Beyond the Horizon. Physics Colloquium - University of Pennsylvania, Philadelphia, United States of America
- 170 (2008). Physics Beyond the Horizon. Colloquium - Perimeter Institute for Theoretical Physics, Waterloo, Canada
- 171 (2008). Observational Evidence for Massive Gravity. PI CITA Joint Cosmology Day 2008, Waterloo, Canada
- 172 (2008). Missing Thermal Energy of the Universe. Astroparticle Workshop - Open Questions from Clusters of Galaxies, Toulouse, France
- 173 (2008). Primordial Non-Gaussianity, Statistics of Collapsed Objects, and the ISW Effect. Origins and Observations of Primordial Non-Gaussianity 2008, Waterloo, Canada
- 174 (2008). Galaxy Clusters as Cosmic Canaries. High Energy Physics Seminar - Syracuse University, Syracuse, United States of America

## Broadcast Interviews

- 2026/07/30 - [3 Physicists on God: Aliens, String Theory, and the Limits of Science](#), Dr Brian Keating, YouTube; with Adam Frank.
- 2026/05/31 - [What Came Before The Big Bang?](#), Apple Finch Pudding, YouTube.
- 2026/04/02 - [The Big Bang May Not Be the Beginning: Dr. Niayesh Afshordi on Time and Nothingness](#), Philosophy Everyday, YouTube.
- 2025/10/09 - [Has Science Just Proved God?](#), Premier Unbelievable? debate with Michel-Yves Bollore, hosted by John Nelson, YouTube.
- 2025/09/16 - [Are We Wrong About the Big Bang?](#), Dr Brian Keating, YouTube.
- 2025/06/12 - [Was There Anything Before Us?](#), Event Horizon, with Phil Halper, YouTube.

- 2025/05/26 - **Mindscape 316: Niayesh Afshordi and Phil Halper on the Big Bang and Before**, Sean Carroll's Mindscape, YouTube.
- 2025/01/19 - **An Astrophysicist Breaks Down Modern Cosmology**, Miles K. Donahue, YouTube.
- 2024/09/27 - Physicists search for Black Hole Bombs, Phil Halper's YouTube Channel, YouTube.com, <https://youtu.be/bJlqWGgt2hw?si=cn9waXQiKvUHIZw3>
- 2024/10/12 - What's inside a black hole? Fuzzballs, Echoes & The Big Bang, Phil Halper's YouTube Channel, YouTube.com, <https://youtu.be/JN9jlg9AdYM?si=c4158UQDt8pRFAfn>
- 2024/05/17 - Did the Universe Create itself ? The PTC model, Before the Big Bang 11, YouTube.com, <https://youtu.be/cLSuM8rKMyk?si=xdh2appUW8wAEi-7>
- 2024/10/12 - The Fine Tuning Argument: the critics strike back, Phil Halper's YouTube channel, YouTube.com, [https://youtu.be/zNH-ZgSpBuQ?si=fahqXSe3\\_F4xgzDv](https://youtu.be/zNH-ZgSpBuQ?si=fahqXSe3_F4xgzDv)
- 2023/03/17 - Echoes from the Abyss: The Mind-Bending Science Behind the Strange Sounds of Black Holes, Time before Space, YouTube.com [https://www.youtube.com/watch?v=d1mcfBip\\_sw&ab\\_channel=TimebeforeSpace](https://www.youtube.com/watch?v=d1mcfBip_sw&ab_channel=TimebeforeSpace)
- 2024/10/12 - Ghazal Geshnizjani & Niayesh Afshordi: Cosmologically Coupled, Conversations at Perimeter Institute, <https://conversationsattheperimeter.podbean.com/>
- 2022/10/09 - A New Model for the Covid Pandemic, Sabine Hossenfelder's channel, YouTube.com, <https://youtu.be/H0oZUpMtJrs?si=TA5cEInNhZRnN5ug>
- 2024/10/12 - Astronomers claim universe has "cosmic glitch", Phil Halper's Youtube Channel <https://youtu.be/GOctev5cUCI?si=LacFQqSuTPKsAUyO>, Youtube
- 2024/05/03 - Fighting COVID-19 with Physics, Interview with the local CTV station in Kitchener-Waterloo, CTV News, Kitchener-Waterloo
- 2024/05/03 - Q&A with the Experts: The problem with Herd Immunity and COVID-19, Mike Farwell Show, 570 News, Kitchener Radio
- 2020/11/20 - Black Hole Genesis, Before the Big Bang, Episode 10, YouTube.com 2019/10/04
- 2021/11/20 - Varying Speed of Light Cosmology, Before the Big Bang, Episode 8, YouTube.com 2019/07/31
- 2020/11/12 - Big Bang and Holography, Science Friday with Ira Flatow, National Public Radio, USA 2017/02/10

- 2017/02/06 - Big Bang, Big Think, Radio Canada International (<https://www.rcinet.ca/en/2017/02/06/the-universe-is-holographic/>)
- 2017/02/06 - Holographic Cosmology, Morning Edition, CBC Kitchener-Waterloo 2017/02/01
- 2017/02/01 - Enigmatic echo from Black Hole, Science@ORF.at, Austrian Broadcasting Corporation 2017/01/01
- 2017/01/01 - Bye Bye Big Bang?, FQXi Podcast, FQXi.org 2013/10/13
- 2013/10/13 - Working at PI, Fresh Air with Mary Ito, CBC Radio 2013/03/16
- 2013/03/16 - Discovery of Higgs Particle at CERN, 60 Minutes, BBC Persian TV 2012/07/04
- 2012/07/04 - Legacy of Stephen Hawking, 60 Minutes, BBC Persian TV 2011/10/04
- 2011/10/04 - (Claimed) Discovery of Superluminal Neutrinos, 60 Minutes, BBC Persian TV 2011/09/24
- 2011/09/24

## Text Interviews

- 2026/07/30 *Science by majority*, text interview by Matthew A. Bell, *Bye Bye Universe*.
- 2024/05/15 Newfound 'glitch' in Einstein's relativity could rewrite the rules of the universe, study suggests, LiveScience.com <https://www.livescience.com/physics-mathematics/newfoundglitch-in-einsteins-relativity-could-rewrite-the-rules-of-the-universe-study-suggests>
- 2024/05/15 A Cosmic Glitch in Gravity, Nautilus Magazine [https://nautil.us/a-cosmic-glitch-ingravity-593171/?\\_sp=0f3668e3-5102-4de7-b03f-fc59eb4ec411.1715821798622](https://nautil.us/a-cosmic-glitch-ingravity-593171/?_sp=0f3668e3-5102-4de7-b03f-fc59eb4ec411.1715821798622)
- 2024/05/01 A 'cosmic glitch' in gravity, Waterloo News <https://uwaterloo.ca/news/media/cosmic-glitchgravity>
- 2024/01/29 Over Gravity's Rainbow: An alternative framework of physics, in which light's speed can vary By Zeeya Merali, John Templeton Foundation Newsletter <https://www.templeton.org/news/over-gravitys-rainbow>
- 2023/02/17 "Dark energy from supermassive black holes? Physicists spar over radical idea", Science News <https://www.science.org/content/article/dark-energy-supermassive-black-holesphysicists-spar-over-radical-idea>
- 2021/09/20 Chiamaka Okoli was a rarity in physics. She challenged norms until her untimely death A Black woman, mother, and a devout Catholic, Okoli was a trailblazer in her field before tragedy struck, Boradview.com <https://broadview.org/chiamaka-okoli/>
- 2021/06/04 Modelling the mysteries of the universe, closer to home Professor Niayesh Afshordi used his knowledge of modelling the mysteries of the Cosmos to start modelling the mysteries of COVID-19, Waterloo News <https://uwaterloo.ca/news/science/modeling-mysteriesuniverse-closer-home>
- 2021/02/08 Echoes from black hole could force us to rethink laws of physics Gravitational echoes may confirm the late Stephen Hawking's quantum black hole hypothesis, Waterloo News, <https://uwaterloo.ca/news/global-impact/echoes-black-hole-could-force-us-rethink-lawsphysics>
- 2020/02/10 Spacetime 'Echoes' From Quantum Black Holes Could Soon Change Physics Forever, VICE Motherboard
- 2020/02/01 Musta aukko sateilee, TAHDET JA AVARUUS Finland
- 2020/01/31 Black holes shouldn't echo, but this one might. Score 1 for Stephen Hawking?, Live Science

2019/10/30 Einstein killed the aether. Now the idea is back to save relativity, New Scientist

2019/06/18 Big Bang BOMBSHELL: Are we living inside a HUGE higher dimensional black hole?, Express.co.uk, <https://www.express.co.uk/news/science/1140059/black-hole-higherdimensional-big-bang-theory-universe-hologram-space-news>

2019/05/24 We've Found Gravitational Waves. What Will We Learn From Them?, Discover Magazine

2019/05/15 Ripple Effect, Discover Magazine

2019/01/16 Crisis at the Heart of Physics, New Scientist

2018/06/20 Total Wipeout, New Scientist

2018/05/14 Echoes of gravitational waves could point to quantum gravity, Physics World

2017/03/29 Echoes of a Black Hole, Aeon.COM

2017/02/04 Could the universe actually be a flat hologram?, Christian Science Monitor

2017/01/31 Lost in space? A brief guide to the 'holographic principle' of the universe, Guardian

2016/12/13 Was Einstein wrong? Controversial theory claims the speed of light is not a constant, Wired Magazine

2016/11/28 Theory Challenging Einstein's View On Speed of Light Could Soon Be Tested, The Guardian

2016/09/12 LIGO black hole echoes hint at general-relativity breakdown, Nature News

2016/02/05 Much Ado around Nothing: The Cosmological non-Constant Problem, Backreaction

2016/01/11 The Sky as a Limit, Perimeter Institute News

2015/02/11 Do Black Hole Firewalls Have Observable Consequences?, Backreaction

2013/09/26 Was Our Universe Born in an Extra-Dimensional Black Hole?, PBS NOVA

2013/09/13 Did a Hyper-Black Hole Spawn the Universe?, Nature News

2013/07/16 Cosmologist Claims Universe May Not Be Expanding, Nature News

2010/07/01 Cosmic Bubble Made Cold Spot in Big Bang Afterglow, New Scientist

2009/03/16 Gravity May Venture Where Matter Fears to Tread, New Scientist

## Publications

Public scholarly record updated 2026-10-02: three new preprints verified directly on arXiv; earlier arXiv, INSPIRE, and ORCID records retained. DOI links point to the published version; arXiv links point to the open manuscript.

### 2026

- <sup>1</sup> Robin Y. Wen, Lukas Tobias Hergt, Niayesh Afshordi, Douglas Scott. Is DESI Seeing Dynamical Dark Energy, or a Cosmic Glitch in Gravity? Preprint (2026). [arXiv:2609.40176](#)
- <sup>2</sup> Zachary Cheslog, Emily Finson, Amanda MacInnis, Neelima Sehgal, Niayesh Afshordi, Simran K. Nerval, Renée Hložek. Current and Future Constraints on the Primordial Power Spectrum. Preprint (2026). [arXiv:2609.35964](#)
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